# Sigmoid / Logistic Regression

From numerical predictions to classes and probabilities

**Worked answers**

Name _____ Date _____

## 01 Binary Classification

* Synthetic practice data Sensor signal $x$ predicts status $y$ Normal $0$, abnormal $1$

**1** Write the possible target values and explain the positive class

$y \in \{$ 0 , 1 $\}$ Positive Class $y =$ 1

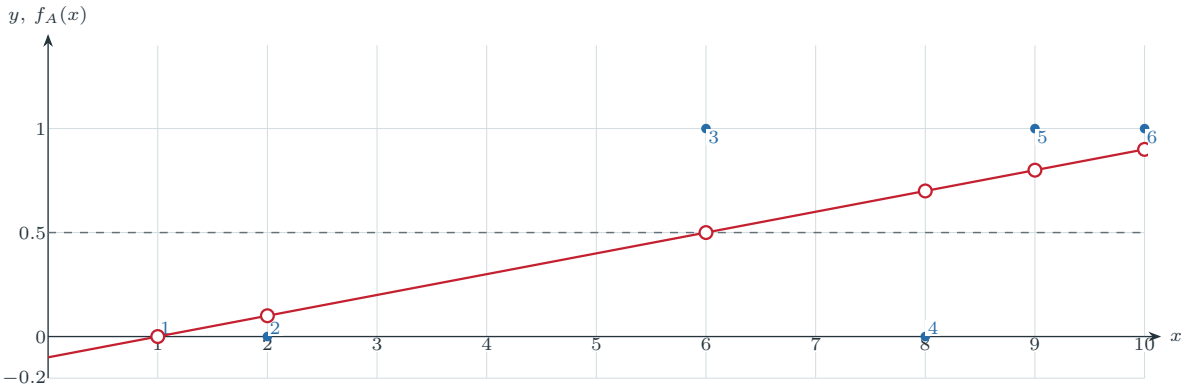
Positive does not mean a desirable state Here label $1$ indicates an abnormal reading

**2** Use line A, fitted to the first six examples, to calculate predictions and classes

$$f_A(x) = 0.1x - 0.1 \qquad \hat{y} = \begin{cases} 1 & f_A(x) \geq 0.5 \\ 0 & f_A(x) < 0.5 \end{cases}$$

| Sensor $i$ | $x$ | $y$ | $f_A(x)$ | $\hat{y}$ |
|---|---|---|---|---|
| 1 | 1 | 0 | $0.1(1) - 0.1 = 0.00$ | 0 |
| 2 | 2 | 0 | $0.1(2) - 0.1 = 0.10$ | 0 |
| 3 | 6 | 1 | $0.1(6) - 0.1 = 0.50$ | 1 |
| 4 | 8 | 0 | $0.1(8) - 0.1 = 0.70$ | 1 |
| 5 | 9 | 1 | $0.1(9) - 0.1 = 0.80$ | 1 |
| 6 | 10 | 1 | $0.1(10) - 0.1 = 0.90$ | 1 |

**3** Plot A's predictions over the actual data and connect them with a straight line




**4** Find $x$ where $f_A(x) = 0.5$ and count correct classifications among the first six examples

$0.1x - 0.1 = 0.5 \quad x =$ $0.6/0.1 = 6$ Correct = 5 /6

## 02 Limits of Linear Regression

Add the seventh sensor example $(x, y) = (27, 1)$ and refit the line using the same MSE

### 5 Compare predictions from A and the new line B at $x = 6$ and $27$

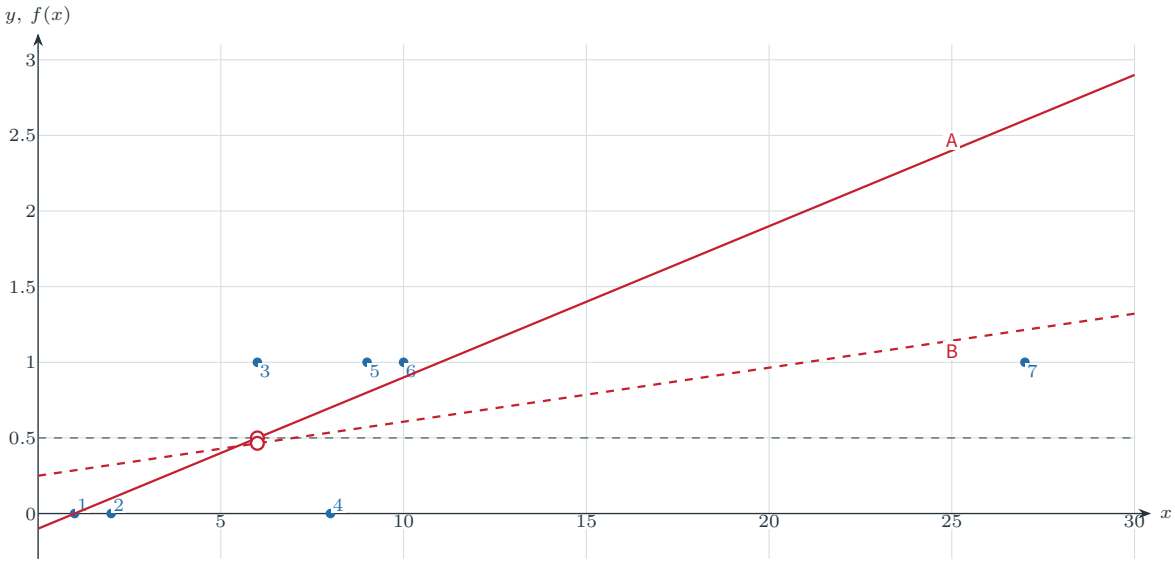
$$f_A(x) = 0.1x - 0.1 \qquad f_B(x) = \frac{x+7}{28} \qquad \hat{y} = 1 \text{ if } f(x) \geq 0.5$$

| $x$ | Target $y$ | $f_A(x)$ | $f_B(x)$ |
|---|---|---|---|
| 6 | 1 | $0.6 - 0.1 = 0.5$ | $\frac{6+7}{28} \approx 0.464$ |
| 27 | 1 | $2.7 - 0.1 = 2.6$ | $\frac{27+7}{28} \approx 1.214$ |

### 6 Calculate A and B at $x = 0, 10, 30$ and draw both lines

| Model | $x = 0$ | $x = 10$ | $x = 30$ |
|---|---|---|---|
| A | $-0.1$ | $1 - 0.1 = 0.9$ | $3 - 0.1 = 2.9$ |
| B | $7/28 = 0.25$ | $17/28 \approx 0.607$ | $37/28 \approx 1.321$ |

* Blue points: actual data A: solid line · B: dashed line




### 7 Explain how the class at $x = 6$ changed and whether all outputs can be interpreted as probabilities

$\hat{y}_A(6) =$ 1 $\hat{y}_B(6) =$ 0

A gives 1 because $0.5 \geq 0.5$; B gives 0 because $0.464 < 0.5$ Adding another $y = 1$ example misclassifies an existing point
A's boundary is $x = 6$; B has $(x + 7)/28 = 0.5$ at $x = 7$ The boundary shifts right
$f_A(27) = 2.6$ and $f_B(27) \approx 1.214$ exceed 1, so they cannot be probabilities
Thresholded linear regression can classify some datasets, but this example exposes its limits

# 03 Sigmoid Function

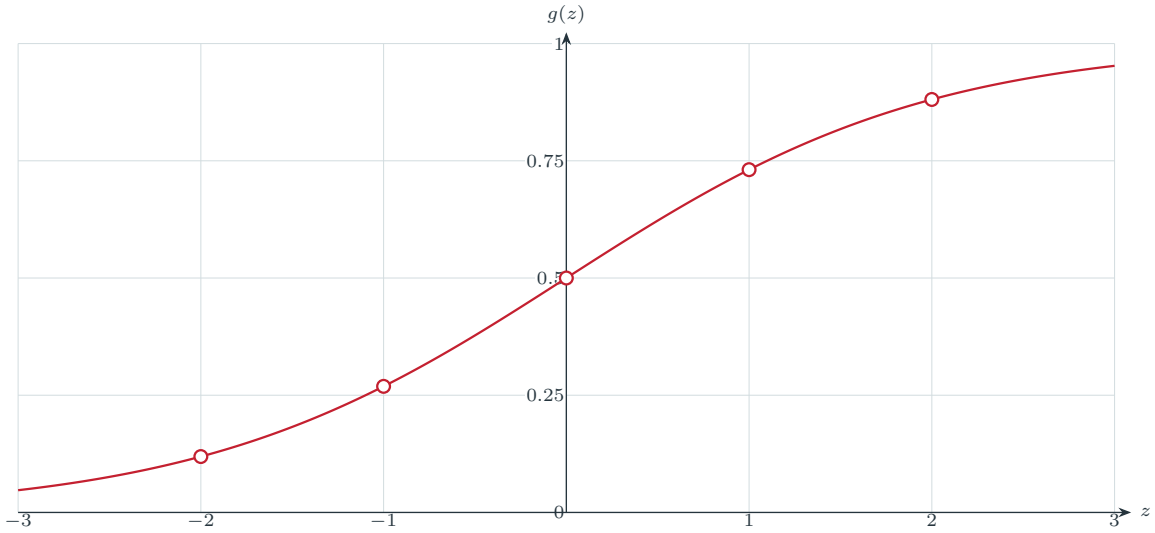
## 8 Use the supplied $e^{-z}$ values to calculate $g(z)$

$$g(z) = \frac{1}{1+e^{-z}} \qquad e \approx 2.718 \qquad e^0 = 1$$

* Record three decimal places A calculator may be used for division and squares

| $z$ | $e^{-z}$ (reference) | $g(z)$ |
|---|---|---|
| $-2$ | $7.389$ | $\frac{1}{1+7.389} \approx 0.119$ |
| $-1$ | $2.718$ | $\frac{1}{1+2.718} \approx 0.269$ |
| $0$ | $1.000$ | $\frac{1}{1+1.000} \approx 0.500$ |
| $1$ | $0.368$ | $\frac{1}{1+0.368} \approx 0.731$ |
| $2$ | $0.135$ | $\frac{1}{1+0.135} \approx 0.881$ |

## 9 Plot $(z, g(z))$ and connect the points smoothly




## 10 Check how the input sign relates to the output and its range

$g(0) = \frac{1}{1+1} = 0.5$ $\qquad z < 0 \Rightarrow g(z) < 0.5$ $\qquad z > 0 \Rightarrow g(z) > 0.5$

Calculate whether equal input changes of $1$ produce equal output changes

$g(0) - g(-1) = 0.500 - 0.269 = 0.231$ $\qquad g(2) - g(1) = 0.881 - 0.731 = 0.150$

Can the output be exactly $0$ or $1$ for a finite $z$?

☐ Yes ☐ No

$e^{-z}$ is always positive, so $1 + e^{-z} > 1$ and $0 < g(z) < 1$
An input change of $1$ gives different output changes: $0.231$ and $0.150$
As $z$ grows, $e^{-z}$ shrinks and $g(z)$ approaches $1$
As $z$ decreases, $e^{-z}$ grows and $g(z)$ approaches $0$ Approaching a value is not the same as reaching it

# 04 Logistic Regression

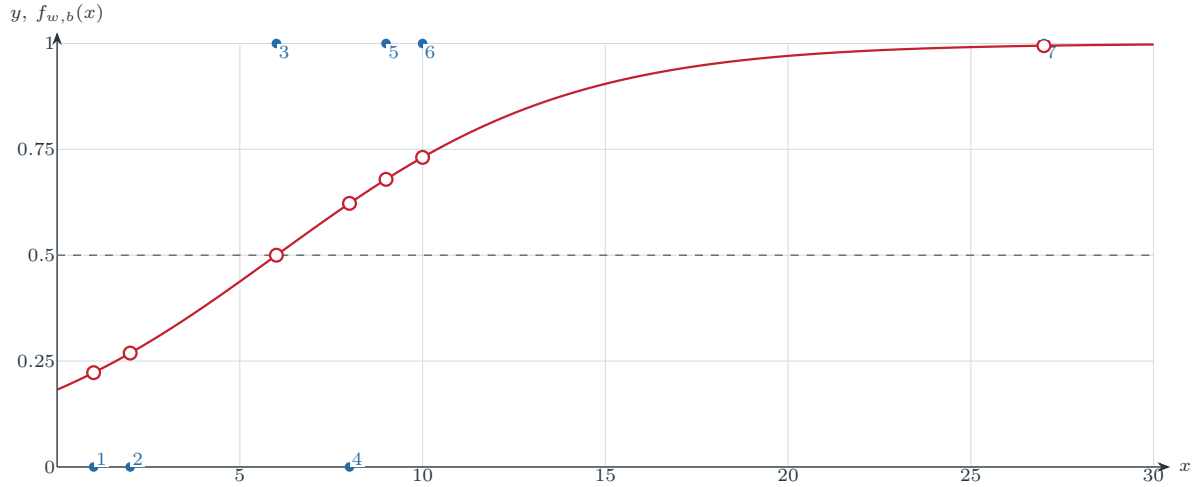
## 11 For all seven examples, calculate score $z$, probability $p$, then class $\hat{y}$

$$z = 0.25x - 1.5 \qquad p = f_{w,b}(x) = g(z) = \frac{1}{1+e^{-z}} \qquad \hat{y} = \begin{cases} 1 & p \geq 0.5 \\ 0 & p < 0.5 \end{cases}$$

$e^{1.25} \approx 3.4903$ $e^{1} \approx 2.7183$ $e^{0} = 1$ $e^{-0.5} \approx 0.6065$
$e^{-0.75} \approx 0.4724$ $e^{-1} \approx 0.3679$ $e^{-5.25} \approx 0.0052$

| Sensor | $x$ | $y$ | $z$ | $p$ | $\hat{y}$ |
|---|---|---|---|---|---|
| 1 | 1 | 0 | $0.25 - 1.5 = -1.25$ | $1/(1+3.4903) \approx 0.223$ | 0 |
| 2 | 2 | 0 | $0.5 - 1.5 = -1$ | $1/(1+2.7183) \approx 0.269$ | 0 |
| 3 | 6 | 1 | $1.5 - 1.5 = 0$ | $1/(1+1.0000) \approx 0.500$ | 1 |
| 4 | 8 | 0 | $2 - 1.5 = 0.5$ | $1/(1+0.6065) \approx 0.622$ | 1 |
| 5 | 9 | 1 | $2.25 - 1.5 = 0.75$ | $1/(1+0.4724) \approx 0.679$ | 1 |
| 6 | 10 | 1 | $2.5 - 1.5 = 1$ | $1/(1+0.3679) \approx 0.731$ | 1 |
| 7 | 27 | 1 | $6.75 - 1.5 = 5.25$ | $1/(1+0.0052) \approx 0.995$ | 1 |

## 12 Plot $(x, p)$ over the actual data and draw the model curve




## 13 Interpret $p \approx 0.731$ at $x = 10$ and calculate the other class probability

$$P(y = 1 \mid x = 10) \approx 0.731 \qquad P(y = 0 \mid x = 10) \approx \quad 1 - 0.731 = 0.269$$

The model predicts about a $73.1\%$ chance that this sensor has $y = 1$ Its actual label is still $0$ or $1$
This does not mean $73.1\%$ accuracy over the dataset Here $0.731 \geq 0.5$, so the predicted class is $1$
At $x = 8$, actual $y = 0$ is misclassified because $p \approx 0.622$ Sigmoid does not remove every exception

# 05 Decision Boundary

**14** Find the input where the preceding model gives $p = 0.5$

$p = g(z) \geq 0.5 \iff z \geq 0 \qquad z = 0.25x - 1.5$

$0.25x - 1.5 = 0 \quad x =$ 6 $\qquad \hat{y} = 1$ if $x \geq 6$

$g(0) = 0.5$, so $0.25x = 1.5$ gives the boundary $x = 6$ The $p \geq 0.5$ rule assigns the boundary to class $1$

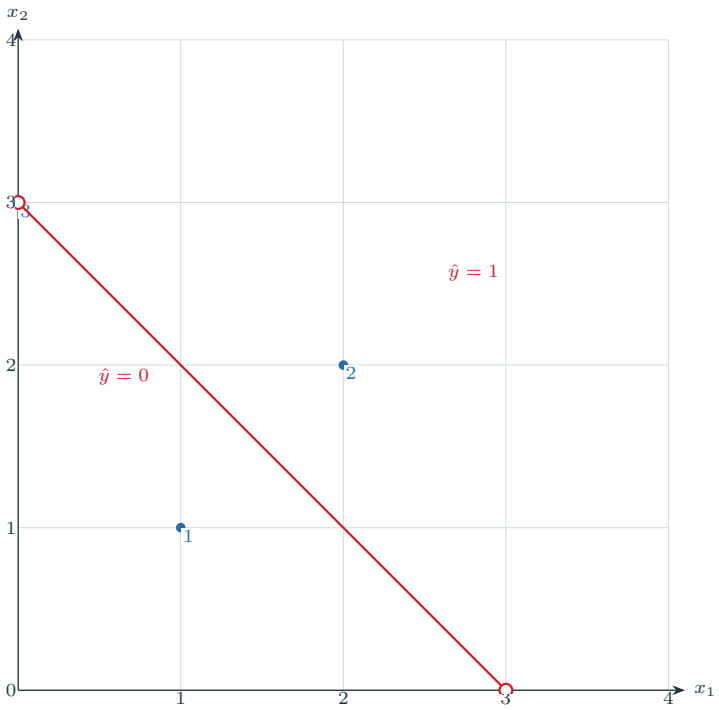
**15** For two inputs, find both boundary intercepts and classify the following points

$z = x_1 + x_2 - 3 \qquad p = g(z)$ Decision Boundary: $z = 0$

$x_1 = 0 \Rightarrow x_2 =$ 3 $\qquad x_2 = 0 \Rightarrow x_1 =$ 3

| $(x_1, x_2)$ | $z$ | $p$ | $\hat{y}$ |
|---|---|---|---|
| $(1, 1)$ | $1 + 1 - 3 = -1$ | $g(-1) \approx 0.269$ | $0$ |
| $(2, 2)$ | $2 + 2 - 3 = 1$ | $g(1) \approx 0.731$ | $1$ |
| $(0, 3)$ | $0 + 3 - 3 = 0$ | $g(0) = 0.5$ | $1$ |

**16** Connect the intercepts and label the regions with $\hat{y} = 0$ and $1$




$x_1 + x_2 = 3$ is the line through $(0, 3)$ and $(3, 0)$
For $x_1 + x_2 < 3$, $z < 0$, so the class is $0$; for $x_1 + x_2 \geq 3$, $z \geq 0$, so it is $1$
The sigmoid plot in $z, g(z)$ is S-shaped, but the boundary $x_1 + x_2 = 3$ in input space is straight

# 06 Linear Model + MSE

* Use all seven examples from here onward  Fix $b = 0$ and vary only $w$ to compare cost shapes

## 17 Substitute $f_w(x) = wx$ into the earlier MSE-based cost and check the expansion

$$J_{\text{linear}}(w) = \frac{1}{2m}\sum_{i=1}^{m}(wx^{(i)} - y^{(i)})^2 = \frac{1015w^2 - 104w + 4}{14} \qquad m = 7$$

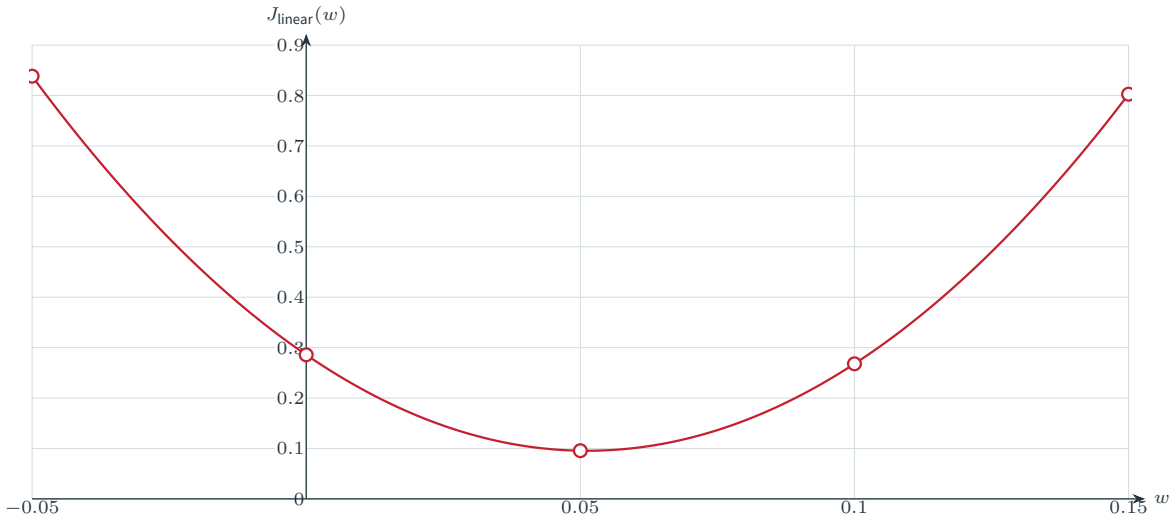
$$\sum_i (x^{(i)})^2 = 1015 \qquad \sum_i x^{(i)}y^{(i)} = 52 \qquad \sum_i (y^{(i)})^2 = 4$$

$$(wx - y)^2 = w^2x^2 - 2wxy + y^2 \quad \text{so} \quad -2\sum_i x^{(i)}y^{(i)} = -2(52) = -104$$

## 18 Calculate the cost at the five values of $w$

| $w$ | Numerator $1015w^2 - 104w + 4$ | $J_{\text{linear}}(w)$ |
|---|---|---|
| $-0.05$ | $2.5375 + 5.2 + 4 = 11.7375$ | $(11.7375)/14 \approx 0.8384$ |
| $0$ | $0 + 0 + 4 = 4$ | $(4)/14 \approx 0.2857$ |
| $0.05$ | $2.5375 - 5.2 + 4 = 1.3375$ | $(1.3375)/14 \approx 0.0955$ |
| $0.1$ | $10.15 - 10.4 + 4 = 3.75$ | $(3.75)/14 \approx 0.2679$ |
| $0.15$ | $22.8375 - 15.6 + 4 = 11.2375$ | $(11.2375)/14 \approx 0.8027$ |

## 19 Plot and connect $(w, J_{\text{linear}}(w))$, then select the lowest-cost $w$ in the table




Selected $w =$ $0.05$  Curve shape = Convex

$0.0955$ is the smallest of the five table costs, so select $w = 0.05$
The true minimum need not be one of the sampled points  The positive $w^2$ coefficient $1015/14 > 0$ gives an upward-opening parabola
The definition here is $1/(2m)$, half the usual MSE; multiplying by $1/2$ does not change the minimizer

# 07 Sigmoid + MSE

## 20 Use the same MSE with $f_w(x) = g(wx)$ to calculate $J_{\text{MSE}}(0)$ and $J_{\text{MSE}}(0.2)$

$$J_{\text{MSE}}(w) = \frac{1}{14}\sum_{i=1}^{7}\left(g(wx^{(i)}) - y^{(i)}\right)^2 \qquad b = 0$$

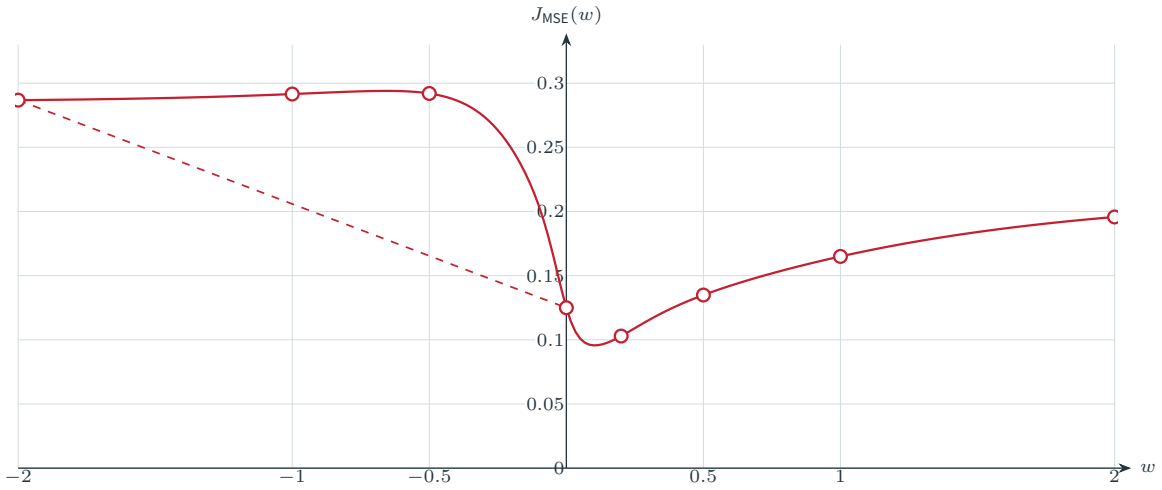
$$w = 0 \Rightarrow g(0x) = 0.5 \quad J_{\text{MSE}}(0) = \frac{7 \times \quad (0.5)^2}{14} = \quad 0.125$$

| $x$ | 1 | 2 | 6 | 8 | 9 | 10 | 27 |
|---|---|---|---|---|---|---|---|
| $y$ | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| $g(0.2x)$ | 0.5498 | 0.5987 | 0.7685 | 0.8320 | 0.8581 | 0.8808 | 0.9955 |
| $(p-y)^2$ | $(0.5498)^2$ 0.30228 | $(0.5987)^2$ 0.35844 | $(-0.2315)^2$ 0.05359 | $(0.8320)^2$ 0.69222 | $(-0.1419)^2$ 0.02014 | $(-0.1192)^2$ 0.01421 | $(-0.0045)^2$ 0.00002 |

$0.5498^2 + 0.5987^2 + (-0.2315)^2 + 0.8320^2 + (-0.1419)^2 + (-0.1192)^2 + (-0.0045)^2$
$\approx 1.440902 \quad J_{\text{MSE}}(0.2) \approx 1.440902/14 = 0.1029$
The table probabilities are rounded to four decimal places Do not round the squared terms during calculation

## 21 Complete the cost table, plot all the points, and connect them

| $w$ | $-2$ | $-1$ | $-0.5$ | 0 | 0.2 | 0.5 | 1 | 2 |
|---|---|---|---|---|---|---|---|---|
| $J_{\text{MSE}}(w)$ | 0.2868 | 0.2915 | 0.2920 | 0.1250 | 0.1029 | 0.1349 | 0.1650 | 0.1957 |




## 22 Join the points at $w = -2$ and $0$ with a line segment and compare its midpoint with the cost at $w = -1$

If convex: $J\left(\frac{a+b}{2}\right) \leq \frac{J(a)+J(b)}{2}$

$$\frac{0.2868 + 0.1250}{2} \approx \quad 0.2059 \qquad J_{\text{MSE}}(-1) = 0.2915 \quad > \quad 0.2059$$

The midpoint cost exceeds the average endpoint cost, violating convexity
As on page 3, equal input changes need not give equal sigmoid output changes; near its extremes the output barely changes
Using these predictions in MSE changes the curvature This dataset gives a nonconvex cost; multiple minima are not inevitable

# 08 Logistic Loss

## 23 Choose the formula for the actual target $y$ and calculate the loss

$$p = f_{w,b}(x) \qquad L(p,y) = \begin{cases} -\log p & y = 1 \\ -\log(1-p) & y = 0 \end{cases}$$

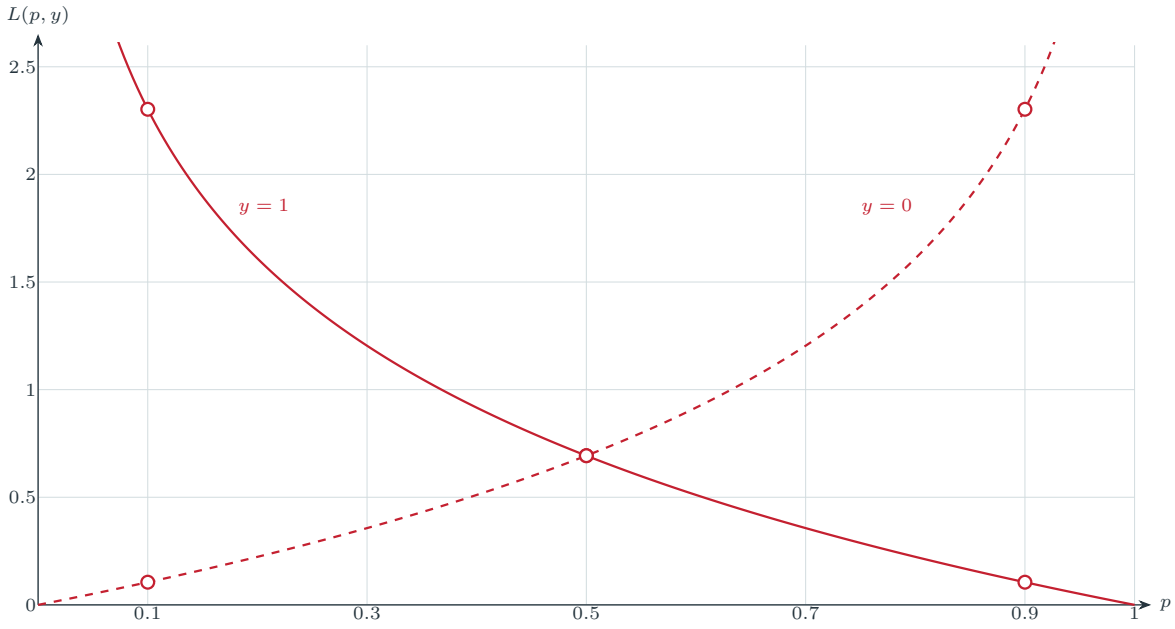
* Loss evaluates one example log denotes the natural logarithm

$\log(0.1) \approx -2.303$ $\log(0.5) \approx -0.693$ $\log(0.9) \approx -0.105$

| $p$ | $L(p,1)$ | $L(p,0)$ |
|---|---|---|
| 0.1 | $-\log(0.1)$ $= -(-2.303) = 2.303$ | $-\log(0.9)$ $= -(-0.105) = 0.105$ |
| 0.5 | $-\log(0.5)$ $= -(-0.693) = 0.693$ | $-\log(0.5)$ $= -(-0.693) = 0.693$ |
| 0.9 | $-\log(0.9)$ $= -(-0.105) = 0.105$ | $-\log(0.1)$ $= -(-2.303) = 2.303$ |

## 24 Plot both sets of loss values on one graph and draw their curves

* $y = 1$: solid line; $y = 0$: dashed line Do not plot infinite values at $p = 0$ or $1$




## 25 For actual target $0$, which prediction has greater loss, $p = 0.1$ or $0.9$? Explain why

At $p = 0.9$, $L = -\log(1 - 0.9) = -\log(0.1) \approx 2.303$
At $p = 0.1$, $L = -\log(1 - 0.1) = -\log(0.9) \approx 0.105$
The more confidently wrong prediction $p = 0.9$ receives a larger loss when the target is $0$
The same probability can therefore be a good or bad prediction depending on the actual label

# 09 Logistic Cost / Convex

## 26 Average the losses over the same seven examples to find $J_{\log}(0)$ and $J_{\log}(0.2)$

$$J_{\log}(w) = \frac{1}{7}\sum_{i=1}^{7} L(g(wx^{(i)}), y^{(i)}) \qquad b = 0 \qquad \log(0.5) \approx -0.6931$$

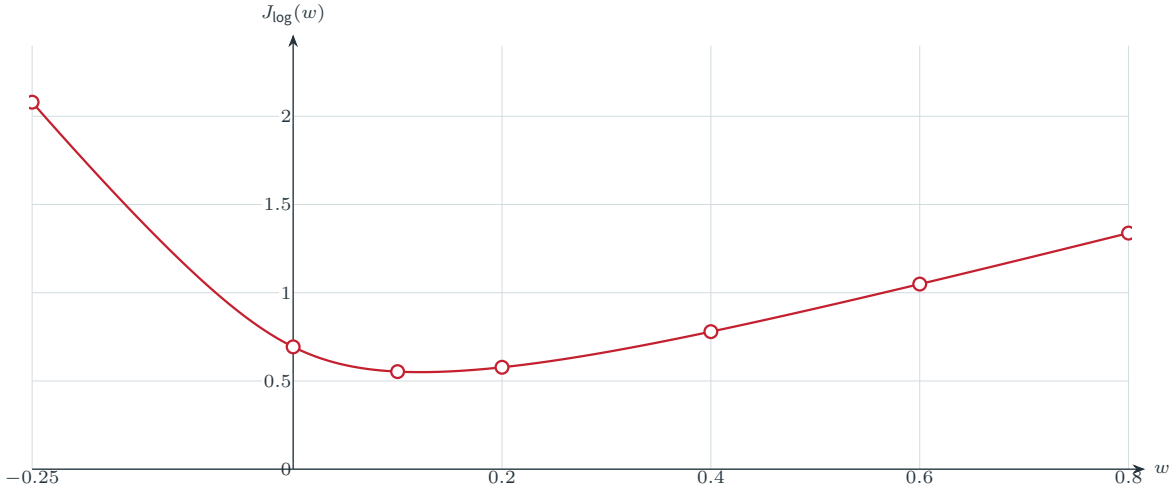
$$J_{\log}(0) = \frac{7 \times \quad 0.6931}{7} \approx \quad 0.6931$$

| $x$ $(w = 0.2)$ | 1 | 2 | 6 | 8 | 9 | 10 | 27 |
|---|---|---|---|---|---|---|---|
| $y$ | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| $\log p$ | $-0.5981$ | $-0.5130$ | $-0.2633$ | $-0.1839$ | $-0.1530$ | $-0.1269$ | $-0.0045$ |
| $\log(1-p)$ | $-0.7981$ | $-0.9130$ | $-1.4633$ | $-1.7839$ | $-1.9530$ | $-2.1269$ | $-5.4045$ |
| $L(p, y)$ | $-(-0.7981)$ $= 0.7981$ | $-(-0.9130)$ $= 0.9130$ | $-(-0.2633)$ $= 0.2633$ | $-(-1.7839)$ $= 1.7839$ | $-(-0.1530)$ $= 0.1530$ | $-(-0.1269)$ $= 0.1269$ | $-(-0.0045)$ $= 0.0045$ |

Use $-\log(1-p)$ for examples 1, 2, 4 ($y = 0$); use $-\log p$ for examples 3, 5, 6, 7 ($y = 1$)
$J_{\log}(0.2) \approx (0.7981 + 0.9130 + 0.2633 + 1.7839 + 0.1530 + 0.1269 + 0.0045)/7$
$= 4.0427/7 \approx 0.5775$ The table logarithms are rounded to four decimal places

## 27 Plot and connect the values below; compare the curve with the preceding Sigmoid + MSE curve

| $w$ | $-0.25$ | 0 | 0.1 | 0.2 | 0.4 | 0.6 | 0.8 |
|---|---|---|---|---|---|---|---|
| $J_{\log}(w)$ | 2.0798 | 0.6931 | 0.5529 | 0.5775 | 0.7794 | 1.0490 | 1.3380 |




## 28 Compare the middle cost with its neighbors at equally spaced $w = 0, 0.2, 0.4$

$$J_{\log}(0.2) \approx 0.5775 \quad < \quad \frac{0.6931 + 0.7794}{2} \approx \quad 0.7363$$

For these three points, the middle cost is below the endpoint average; they do not violate the test from page 7
Three points do not prove global convexity With fixed features, the logistic-loss cost is convex in $w, b$
Sigmoid alone does not determine the cost shape The choice of loss matters even for identical predictions
The horizontal axis on page 8 is probability $p$; here it is parameter $w$ Distinguish a loss curve from a parameter cost curve